

Education

Birla Institute of Technology and Science, Pilani

B.E. Electronics and Communication Engineering

2023 – Present

CGPA: 9.30/10

Research and Publications

Inverse Design of Realizable Metasurface-Based Absorbers Using Improved Conditioning and Diversity-Enhanced Progressively Growing GANs

Scientific Reports (2026), published 18 September 2026. doi:10.1038/s41598-026-69259-y | arXiv:2606.05849

- Co-authored a fabrication-aware inverse-design framework combining continuous FiLM conditioning, electromagnetic-surrogate guidance, and diversity regularisation to generate absorber designs across 2–18 GHz.
- Reported 0.0052 average MSE, 0.8730 diversity score, 0.8533 band-alignment accuracy, and an 89.57% valid-design rate.

Experience

D. E. Shaw

Jun – Jul 2026

Technology Development Intern, Quant Systems Development

- Evaluated identity, firewall access, DNS, proxy routing, and TLS requirements for extending service discovery into a network-restricted domain.
- Developed an eBPF program to integrate a virtual machine’s Layer 3 network space into the firm’s grid network for userspace testing of network technologies.

Selected Engineering Projects

openEMS-CUDA: Batched GPU Electromagnetic Simulation

2026

Independent systems and GPU engineering

C++, CUDA, Python

- Built a batched CUDA FDTD engine for openEMS sweeps, using fused kernels, CUDA Graph replay, batched port monitors, and Python orchestration for ragged grids and independent early stopping.
- Matched the reference GPU engine bit-for-bit on E-fields; a 16-configuration dipole benchmark reached 15.5 simulations/s with at most 2.2×10^{-7} relative error in the I time series. Code

Making Computation Fit: Edge AI and CXL-Backed MoE

2026

Hardware and systems research

SystemVerilog, FPGA, CXL, Python

- Built a six-horizon streaming power-forecasting pipeline and optimised its TCN for Zynq-7020; synthesis reduced LUT use by 41% and flip-flops by 67% (Vivado 2019.2; 28 BRAM36 blocks, 48 DSPs).
- Evaluated CXL memory expansion for Qwen3-30B-A3B serving: simulation placed a 56-GiB model beyond 48 GiB aggregate HBM, but incurred $27.05\times$ mean TTFT and $213.05\times$ mean TPOT versus ideal HBM.

Nebula GPU Interconnect Design Competition

Aug – Sep 2025

Second Place, Hardware Design

SystemVerilog, NoC

- Designed a network-on-chip for up to 64 GPUs with custom deadlock-free arbitration and AXI/CHI-compatible interconnect modules.
- Tested shared GPU-training traffic patterns; placed second in the hardware-design category of the Astera Labs Nebula competition.

Team Anant Satellite Simulation (sat-sim)

2025 – 2026

Satellite Systems Engineering

Python, orbital dynamics

- Refactored Team Anant's orbit-propagation sandbox into a modular spacecraft-analysis architecture; developed solar-power and Monte Carlo analysis workflows.
- Contributed TT&C link/contact analysis and magnetorquer detumbling features to a simulator covering orbital perturbations and spacecraft subsystems. Code

Gradient-Based Antenna Array Optimisation

Aug – Dec 2025

Research project

PyTorch, EM solvers

- Applied gradient-based optimisation to array-factor models for wireless-power transfer, directivity, and beam steering; compared against differential evolution, genetic algorithms, and particle swarm optimisation. Code

Algorithmic Trading Competition Platform

Mar – Apr 2026

Tech-fest competition backend

Go, Python, Protocol Buffers

- Built a Go order book and matching engine with concurrent Python-strategy evaluation over Protocol Buffers, supporting 200+ strategies per one-second market tick. Code

Selected DSP and Campus Software

2023 – 2026

Research, club, and event engineering

Python, Go, Next.js

- Refactored modules and tests for a chaotic-encryption audio watermarking system using three-level Haar DWT and DTMT; achieved 0.993 correlation under 20 dB Gaussian noise and 0.998 after 128 kbps MP3 on the tested track. Code
- Built a Go/Gin/SQLC authentication service for the BITS Postman recruitment portal, including Google OAuth sessions, token verification, and club-head login.
- Built Team Anant's website, contributed to its Flask ERP (projects, inventory, finance, roles), and created the Qiskit Fall Fest 2026 site for an 8-day, 16-session, 27-hour programme.

Leadership and Campus Involvement

Systems Engineer - Team Anant, BITS Pilani satellite club

May 2024 – Present

- Coordinate subsystem interfaces for a 3U CubeSat mission.
- Work across TT&C link budgets and modulation schemes, power budgeting, OBC state transitions, and ADCS state estimation with Kalman filtering; consider magnetorquer actuation and payload constraints in mission design.

Cultural Fest Coordinator - Matrix, The Film and Literature Club

May – Dec 2025

- Organised the book fair and ran film screenings and quizzes as part of the cultural fest.

IEEE BPSC (AI/ML Vertical); Coding Club (competitive programming); The Debating Society.

Technical Skills

Programming: Go, Python, C++, SystemVerilog

Systems: Linux, networking, eBPF, concurrency, Protocol Buffers

Hardware: RTL design, NoC and GPU interconnects

Machine Learning and Optimisation: PyTorch, GANs, diffusion models, gradient-based optimisation

Selected Coursework

Deep Learning; Quantum Informatics and Computing; Special Relativity; Group Theory; Topology; Cryptography.

Additional Information

Placement offer: Graviton Research Capital.